4.6 Change of basis

V has bases $\mathcal{B} = \{b_1, \dots, b_n\}$ and

$\mathcal{W} = \{w_1, \dots, w_n\}$

$$\begin{tikzcd}
 & V \arrow[dl, leftrightarrow, "{[\cdot]_{\mathcal{W}}}"'] \arrow[dr, leftrightarrow, "{[\cdot]_{\mathcal{B}}}"] & \\
\mathbb{R}^n & & \mathbb{R}^n \arrow[ll, "P_{\mathcal{W} \leftarrow \mathcal{B}}"]
\end{tikzcd}$$

i.e. $$[v]_{\mathcal{W}} = P_{\mathcal{W} \leftarrow \mathcal{B}} \, [v]_{\mathcal{B}}$$

Thm 15 $$P_{\mathcal{W} \leftarrow \mathcal{B}} = \begin{bmatrix} [b_1]_{\mathcal{W}} & \cdots & [b_n]_{\mathcal{W}} \end{bmatrix} \in M_{n \times n}$$

(unique).

Remarks The matrix $P_{\mathcal{W} \leftarrow \mathcal{B}}$ is invertible since it comes from the composition of two invertible maps $[\cdot]_{\mathcal{B}}^{-1}$ and $[\cdot]_{\mathcal{W}}$.

Let $\left(P_{\mathcal{W} \leftarrow \mathcal{B}}\right)^{-1}$ be its inverse matrix.

Multiplying $[v]_{\mathcal{W}} = P_{\mathcal{W} \leftarrow \mathcal{B}} \, [v]_{\mathcal{B}}$ on the left by $\left(P_{\mathcal{W} \leftarrow \mathcal{B}}\right)^{-1}$ we obtain

$$\left(\underset{\mathcal{W}\leftarrow\mathcal{B}}{P}\right)^{-1}[v]_{\mathcal{W}} = \cancel{\left(\underset{\mathcal{W}\leftarrow\mathcal{B}}{P}\right)^{-1}}\cancel{\left(\underset{\mathcal{W}\leftarrow\mathcal{B}}{P}\right)}[v]_{\mathcal{B}}$$

$$= I\,[v]_{\mathcal{B}}$$

$$= [v]_{\mathcal{B}}$$

$$\Rightarrow \quad [v]_{\mathcal{B}} = \left(\underset{\mathcal{W}\leftarrow\mathcal{B}}{P}\right)^{-1}[v]_{\mathcal{W}}$$

By the uniqueness in Thm 15

$$\left(\underset{\mathcal{W}\leftarrow\mathcal{B}}{P}\right)^{-1} = \underset{\mathcal{B}\leftarrow\mathcal{W}}{P}$$

Example Take $V = \mathbb{P}_2$, $\mathcal{B} = \{1, x, x^2\}$

and $\mathcal{W} = \{1, 1+x, 1+x+x^2\}$ (Exercise: $\mathcal{W}$ is a basis)

Let's compute $\underset{\mathcal{W}\leftarrow\mathcal{B}}{P}$.

$$1 = 1(1) + 0(1+x) + 0(1+x+x^2)$$

$$\Rightarrow [1]_{\mathcal{W}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$x = (-1)(1) + (1)(1+x) + (0)(1+x+x^2)$$

$$\Rightarrow [x]_{\mathcal{W}} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$

$$x^2 = (0)(1) + (-1)(1+x) + (1)(1+x+x^2)$$

$$\Rightarrow [x^2]_{\mathcal{W}} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\underset{\mathcal{W} \leftarrow B}{P} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$[p]_{\mathcal{W}} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} [p]_B \quad \forall p \in \mathbb{P}_2$$

Take $p(x) = 3 + x - 2x^2$.

Then $[p]_B = \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix}$

$$[p]_{\mathcal{W}} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ -2 \end{bmatrix}$$

Exercise take $p(x) = a_0 + a_1 x + a_1 x^2$

and compute

$$[p]_{\mathcal{W}} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix}$$

Example

Suppose we know $[v]_B$, $\underset{\mathcal{W} \leftarrow B}{P}$. Then

$[v]_{\mathcal{W}} = \underset{\mathcal{W} \leftarrow B}{P} [v]_B$. If we also know what $\mathcal{W}$ is then we can compute v itself.

Take $V = \mathbb{R}^2$ and $\mathcal{W} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix} \right\}$, $[v]_B = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

and $\underset{\mathcal{W} \leftarrow B}{P} = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$.

Then $[v]_{\mathcal{B}} = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \end{bmatrix}$.

This means $v = 3 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 3 \\ 3 \end{bmatrix}$

5.1 Eigenvectors and eigenvalues

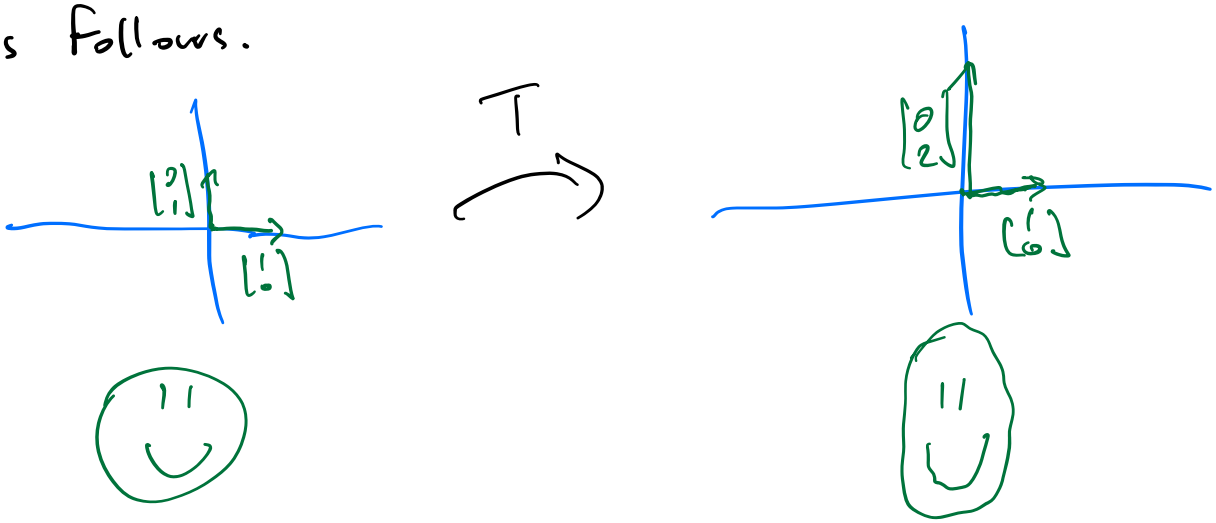
In this section $A \in M_{n \times n}$ and $T : \mathbb{R}^n \to \mathbb{R}^n$ is the lin. transformation defined by $T(v) = Av \quad \forall v \in \mathbb{R}^n$.

Example Take $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$

Then $T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix} = (1) \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

and $T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

What is T doing to $\mathbb{R}^2$? We can visualize this as follows.

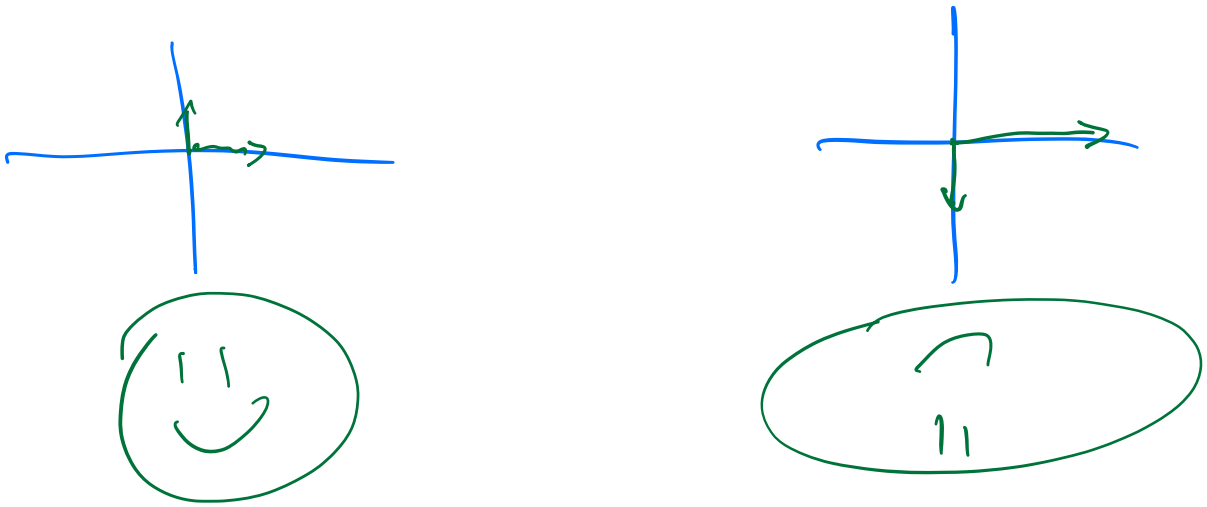


Example

Take $A = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$

Then $T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = 2\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$T\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right) = -\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

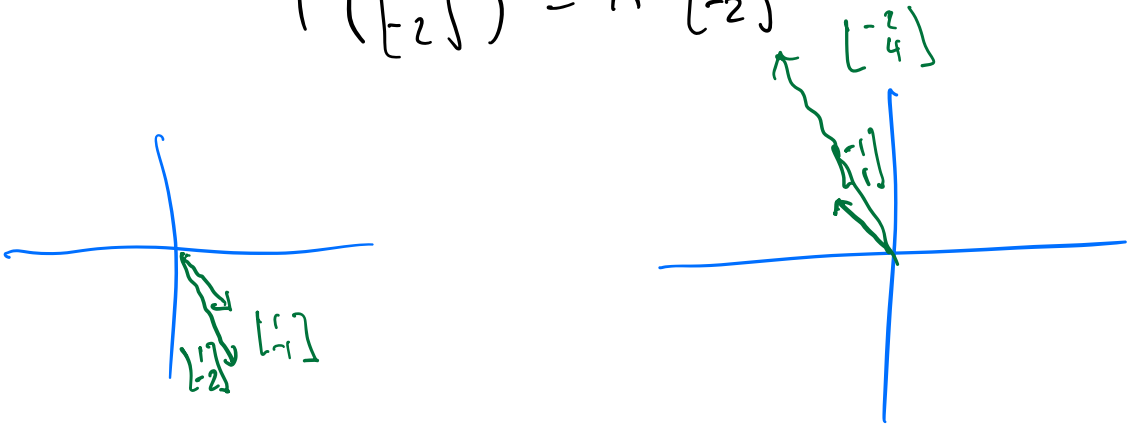


Example Take $A = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix}$. Then

$T\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) \neq \lambda \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ for any $\lambda \in \mathbb{R}$.

However $T\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = A\begin{bmatrix} 1 \\ -1 \end{bmatrix} = \cdots = (-1)\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$T\left(\begin{bmatrix} 1 \\ -2 \end{bmatrix}\right) = A\begin{bmatrix} 1 \\ -2 \end{bmatrix} = \cdots = (-2)\begin{bmatrix} 1 \\ -2 \end{bmatrix}$.



Dream Express T as a matrix that scales basis vectors with some constants.

Definition

An <u>eigenvector</u> of A is a <u>nonzero</u> vector $v \in \mathbb{R}^n$ such that $Av = \lambda v$ for some $\lambda \in \mathbb{R}$.
In this case λ is called the <u>eigenvalue</u> (of v).

Theorem

Suppose $\lambda \in \mathbb{R}$ is an eigenvalue of A. Then $v \in \mathbb{R}^n$ is an eigenvector for λ if and only if

(i) $v \neq 0$
(ii) $v \in \mathrm{Nul}(A - \lambda I)$

Proof. $v \in \mathbb{R}^n$ is an eigenvector for λ

$\iff$ $Av = \lambda v$ and $v \neq 0$

$\iff$ $Av = \lambda I v$ and $v \neq 0$

$\iff$ $Av - \lambda I v = 0$ and $v \neq 0$

$\iff$ $(A - \lambda I)v = 0$ and $v \neq 0$

$\iff$ $v \in \mathrm{Nul}(A - \lambda I)$ and $v \neq 0$.

Definition The λ-<u>eigenspace</u> of A is $\mathrm{Nul}(A - \lambda I)$

Remarks – Recall that any null space is a subspace (of $\mathbb{R}^n$)

The λ-eigenspace = set of all eigenvectors for λ together with 0.

Q Why is 0 not allowed as an eigenvector?

A. $A(0) = 0 = \lambda 0$. for <u>all</u> $\lambda \in \mathbb{R}$ and for all $A \in M_{n \times n}$

So 0 is giving no information about a particular matrix A.

Example Take $A = \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix}$. Take for granted

that 2 is an eigenvalue of A.

The 2-eigenspace $= \mathrm{Nul}(A - 2I)$

$$= \begin{bmatrix} 4 & -1 & 6 \\ 2 & 1 & 6 \\ 2 & -1 & 8 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & -1 & 6 \\ 2 & -1 & 6 \\ 2 & -1 & 6 \end{bmatrix}$$

The REF of $A - 2I$ is $\begin{bmatrix} 1 & -1/2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

The null space $\mathrm{Nul}(A - 2I) = \left\{ \begin{bmatrix} 1/2\, c_2 - 3c_3 \\ c_2 \\ c_3 \end{bmatrix} : c_2, c_3 \in \mathbb{R} \right\}$

$$= \left\{ c_2 \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix} + c_3 \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} : c_2, c_3 \in \mathbb{R} \right\}$$

$$= \operatorname{Span} \left\{ \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$$

By a theorem $\left\{ \begin{bmatrix} 1/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix} \right\}$ is a basis for the 2-eigenspace.